Project Report

Course: Python Essentials
Course Code: CSE1021
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1. Problem Statement:

Sleep is one of the most critical components of human health and productivity. Despite this, most students and working individuals do not consistently track their sleep habits. When sleep patterns are poor, the consequences include reduced concentration, weakened immunity, and long-term health risks.

Most sleep-tracking solutions available today require smartphone apps, smartwatches, or subscription services. These are either expensive or overly complex for someone who simply wants to log and reflect on their sleep.

This project solves that problem with a lightweight, command-line Python application that anyone can run on a basic computer; no internet, no app, no hardware required.

2. Why This Problem Matters?

During the course of college life, students often compromise sleep due to academic pressure, irregular schedules, or lack of awareness. Without tracking, it is nearly impossible to identify poor sleep patterns. A simple logging tool can bring awareness and encourage healthier habits.

This problem is real and personally relevant. Many students sleep fewer than 6 hours on weekdays without realizing the pattern until they see data. The goal of this project was to make that data visible in the simplest way possible.

3. Solution Overview

The Sleep Tracker is a command-line Python application that allows a user to:

- Log each night's sleep with date, hours, quality rating, and optional notes.
- View all previous sleep entries in a formatted table.
- Generate a weekly report showing average sleep, best/worst night, and a health tip.
- Persist data in a CSV file that retains all entries across sessions

The program is menu-driven and fully interactive via the terminal.

4. Python Concepts Applied

This project directly applies the following concepts covered in the Python Essentials course:

FUNCTIONS: Separate functions for each feature (log, view, report).

LOOPS: ‘While -True’ main loop for the menu; ‘for’ loops to read CSV rows.

CONDITIONALS: 'if/elseif/else' for menu choices and health tip logic.

FILE HANDLING: 'csv' module to read from and write to a CSV file.

LISTS & DICTS: Storing and processing rows of sleep data

STRING FORMATTING: f-strings for clean, aligned output

EXCEPTION HANDLING: 'try/except' for invalid numeric input.

‘DATE TIME’ MODULE: Parsing dates, calculating 7-day windows for weekly report.

‘OS’ MODULE: Checking if the CSV file already exists before creating it

5. Key Design Decisions

CSV over a database: A CSV file was chosen for data storage because it requires no third-party libraries, is human-readable, and is sufficient for the scale of this project. Any user can open the file in Excel or Google Sheets to view their data directly.

Quality rating: A subjective quality score was included because sleep duration alone does not capture the full picture. Someone may sleep 8 hours but feel unrested. The quality metric adds a second dimension to the data.

7-day rolling window for reports: Rather than requiring the user to specify a date range, the report automatically covers the last 7 days. This makes it effortless to use while still being meaningful.

Health tip logic: A simple rule-based tip is generated based on average sleep hours. This makes the output actionable rather than just informational.

6. Challenges Faced:

Date handling: One of the initial challenges was parsing and comparing dates correctly. Python's ‘datetime’ module required careful handling to convert CSV strings back into ‘date’ objects for comparison.

Input validation: Ensuring that the user cannot enter text where numbers are expected required wrapping input in ‘try/except ValueError’ blocks. This was a good practical exercise in defensive programming.

Edge cases in the report: When no data existed for the last 7 days, the program needed to handle that gracefully without crashing. This was solved with an early return and a friendly message.

7. What I Learned?

This project reinforced several important lessons:

Thinking in functions: Breaking a program into small, single-purpose functions made the code much easier to read and debug. Each feature (log, view, report) became its own contained unit.

File I/O is powerful: Persisting data to a CSV file was surprisingly straightforward with Python's built-in `csv` module, and it made the application feel truly useful rather than a one-time script.

User experience matters even in CLI tools: Small decisions like showing a confirmation message after logging, using column headers in the view, or providing a health tip make the program feel polished and thoughtful.

Error handling is essential: Without input validation, the program would crash on bad input. Handling exceptions properly was the difference between a fragile script and a reliable tool.

8. Sample Output:

```
========================================
    Sleep Tracker - Stay Rested!
========================================

What would you like to do?
1. Log tonight's sleep
2. View all logs
3. Weekly sleep report
4. Exit

Enter choice (1-4): 3

--- Weekly Sleep Report (Last 7 Days) ---
Days Logged: 5
Average Sleep: 6.8 hours/night
```

```
Average Quality: 7.2/10
Best Night: 2026-03-28 (8.5 hrs)
Worst Night: 2026-03-26 (5.0 hrs)

 Health Tip:
   You're slightly below the recommended 7-8 hours. Try
sleeping 30 mins earlier.
```

9. Conclusion

The Sleep Tracker project successfully demonstrates the practical application of Python fundamentals to solve a real, everyday problem. The solution is purposeful, well-organized, and usable by anyone without any technical background.

The project also reinforced my confidence in building end-to-end programs â€” from taking user input, to processing and storing data, to generating meaningful output. I look forward to extending this in the future with features like graphical charts or a mobile interface.

10. References

* Python Official Documentation: https://docs.python.org/3/
* Python `csv` module: https://docs.python.org/3/library/csv.html
* Python `datetime` module: https://docs.python.org/3/library/datetime.html
* National Sleep Foundation sleep guidelines: https://www.sleepfoundation.org